

UNIVERSITY LIBRARY MANAGEMENT SYSTEM

Assignment Report — Database Systems

Week 11 | Undergraduate Level | Total Marks: 20

Course	Database Systems
Level	Undergraduate (UG)
Total Marks	20 Marks
Submission	Week 11
Tools Used	MySQL · Node.js / Express · React
System	University Library Management System

Prepared for submission as part of the Database Systems module assessment.

Scenario Overview

The University Library Management System is a relational database application designed to manage and track the core operations of a university library. It stores structured information about enrolled students, the available book catalogue, and all borrowing transactions. The system empowers librarians and administrators to retrieve data efficiently, process new loans, and generate meaningful reports — replacing slow and error-prone manual record-keeping.

This report demonstrates the design and implementation of SQL queries, data-entry forms, and management reports as required by the assignment. All work is supported by a live full-stack application built using MySQL, Node.js/Express, and React.

Database Structure

Three relational tables form the core of the system, with sample data as shown below.

Table 1 — Students

StudentID	StudentName	Program	Year	Email
S101	Ali Khan	BSCS	2	ali.khan@uni.edu
S102	Sara Ahmed	BBA	1	sara.ahmed@uni.edu
S103	John Lee	BSIT	3	john.lee@uni.edu

Table 2 — Books

BookID	Title	Author	Category
B201	Database Fundamentals	Elmasri	IT
B202	Business Management	Robbins	Business
B203	Web Development	Jon Duckett	IT

Table 3 — Borrow

BorrowID	StudentID	BookID	BorrowDate	ReturnDate	Status
BR01	S101	B201	01-03-2026	10-03-2026	Returned
BR02	S103	B203	02-03-2026	12-03-2026	Returned

Question 1 — Query Design (8 Marks)

Queries are the backbone of any database system. They allow users to retrieve precisely the data they need from one or more tables using structured SQL statements. The following three queries were designed and implemented for the University Library Management System.

Q1-a) Student Name, Book Title, and Borrow Date

This query retrieves the full list of borrowing transactions by joining all three tables. It shows who borrowed which book and when, providing a complete overview of loan activity.

```
SELECT Students.StudentName,  
       Books.Title AS BookTitle,  
       Borrow.BorrowDate  
FROM Borrow  
JOIN Students ON Borrow.StudentID = Students.StudentID  
JOIN Books ON Borrow.BookID = Books.BookID;
```

Query Result

StudentName	BookTitle	BorrowDate
Ali Khan	Database Fundamentals	2026-03-01
John Lee	Web Development	2026-03-02

How it works: The JOIN operations link the Borrow table to both Students and Books using their shared foreign keys (StudentID and BookID). This produces a single merged result set that avoids the need to cross-reference tables manually.

Q1-b) All Books in the IT Category

This query filters the Books table to return only those titles categorised as IT. It is a straightforward single-table query using the WHERE clause — but highly efficient when the Category column is indexed by the DBMS.

```
SELECT BookID,  
       Title,  
       Author,  
       Category
```

```
FROM Books
WHERE Category = 'IT';
```

Query Result

BookID	Title	Author	Category
B201	Database Fundamentals	Elmasri	IT
B203	Web Development	Jon Duckett	IT

How it works: The WHERE Category = 'IT' clause acts as a filter. The DBMS checks every row and returns only those matching the condition — B201 and B203. B202 (Business Management) is correctly excluded.

Q1-c) Students Who Borrowed After 1 March 2026

This query identifies students who borrowed books strictly after the 1st of March 2026. It is useful for periodic reviews — for example, verifying that loan records are current or identifying late-semester borrowing activity.

```
SELECT Students.StudentName,
Borrow.BorrowDate,
Books.Title AS BookTitle
FROM Borrow
JOIN Students ON Borrow.StudentID = Students.StudentID
JOIN Books ON Borrow.BookID = Books.BookID
WHERE Borrow.BorrowDate > '2026-03-01';
```

Query Result

StudentName	BorrowDate	BookTitle
John Lee	2026-03-02	Web Development

How it works: The strict greater-than operator (>) on BorrowDate excludes any loan made on 01-03-2026 itself. Only BR02 (John Lee, 02-03-2026) satisfies this condition. Ali Khan borrowed on exactly 01-03-2026 and is correctly excluded.

Q1-d) Analysis — Queries vs Manual Searching

A database query and a manual table search ultimately aim to do the same thing — find specific information. However, their performance, reliability, and scalability are worlds apart. The comparison table

below highlights the key differences.

Criterion	SQL Query	Manual Searching
Speed	Milliseconds via indexed lookup ($O \log n$)	Minutes for large tables ($O n$ linear scan)
Accuracy	100% consistent — same query, same result	Human error-prone; records can be missed
Multi-table	JOIN combines multiple tables in one pass	Requires manual cross-referencing of tables
Filtering	WHERE, HAVING, BETWEEN — precise and flexible	Must visually scan every row for conditions
Reusability	Save once; run instantly whenever needed	Must repeat the full process each time
Aggregation	COUNT, SUM, AVG computed automatically	Requires manual tallying; very error-prone

In practice, a SQL query against a table of 10,000 borrow records will return results in under 100 milliseconds, while a librarian manually scanning the same records would take hours. Moreover, queries are deterministic — two staff members running the same query always get identical results, which is not the case with manual review. As data grows, the performance gap widens dramatically, making SQL the only viable approach for any production library system.

Question 2 — Form Creation (6 Marks)

Forms serve as the primary interface between users and the database. They replace raw SQL INSERT statements with guided, validated screens that any staff member can use without technical knowledge. Two forms were designed for the system.

Q2-a) Student Data Entry Form

The Student Entry Form allows the library administrator to register new students. Every field maps directly to a column in the Students table, and the form enforces validation rules before any INSERT operation is executed.

Field Label	Input Type	Validation Rule	DB Column
Student ID	Text input	Required, unique (PRIMARY KEY)	StudentID VARCHAR(10)
Full Name	Text input	Required, max 100 characters	StudentName VARCHAR(100)
Program	Dropdown (BSCS/BBA/...)	Required, predefined list only	Program VARCHAR(50)
Year	Dropdown (1 to 4)	Required, integer 1–4 only	Year INT
Email	Email input	Optional, valid email format	Email VARCHAR(100)

Generated SQL on Form Submission

```
INSERT INTO Students (StudentID, StudentName, Program, Year, Email)
VALUES ('S104', 'Jane Smith', 'BSCS', 2, 'jane.smith@uni.edu');
```

The form uses a React frontend with client-side validation. If any required field is left blank, a red error message appears immediately below the field without submitting the form. On success, a green confirmation alert is shown and the student list refreshes. API endpoint: POST /students.

Q2-b) Book Borrowing Form

The Borrow Form enables librarians to record a new book loan in seconds. Rather than requiring staff to know valid IDs from memory, the form dynamically loads dropdown menus populated from the live database — ensuring only valid foreign keys can be submitted.

Field Label	Input Type	Validation Rule	DB Column
Borrow ID	Text input	Required, unique (PRIMARY KEY)	BorrowID VARCHAR(10)
Student	Dropdown (from database)	Required, valid FK to Students	StudentID VARCHAR(10)
Book	Dropdown (from database)	Required, valid FK to Books	BookID VARCHAR(10)
Borrow Date	Date picker	Required, valid calendar date	BorrowDate DATE
Return Date	Date picker	Optional — can be updated later	ReturnDate DATE

Generated SQL on Form Submission

```
INSERT INTO Borrow (BorrowID, StudentID, BookID, BorrowDate, ReturnDate, Status)
VALUES ('BR03', 'S102', 'B202', '2026-03-05', '2026-03-15', 'Active');
```

A transaction preview panel appears below the dropdowns as the librarian makes selections, showing the student name and book title before final submission. This reduces accidental selection errors. API endpoint: POST /borrow.

Q2-c) How Forms Improve Data Accuracy and Usability

Forms are one of the most important features of any database application. Without them, data entry requires staff to write SQL directly — a process that is slow, technically demanding, and very easy to get wrong. Forms bridge the gap between the database engine and everyday users.

DATA ACCURACY IMPROVEMENTS:

1. Input validation catches errors before they reach the database. Required fields cannot be skipped, numeric fields reject text, and email fields enforce proper format.
2. Dropdown menus for foreign key fields (Student, Book) restrict choices to values that already exist in the database — preventing referential integrity violations.
3. Feedback messages — success alerts in green, errors in red — confirm each action completed correctly, reducing uncertainty for the user.

USABILITY IMPROVEMENTS:

1. Non-technical staff such as library assistants can operate the system without any SQL knowledge. Labels, placeholders, and tooltips guide them through each field.
2. The borrow form auto-populates student and book options from the live database, so librarians never need to memorise ID codes.
3. A transaction preview panel shows the librarian a human-readable summary before they confirm — mimicking a checkout flow that feels natural and familiar.

Question 3 — Report Generation (6 Marks)

Reports transform raw database records into organised, meaningful summaries that support decision-making. Unlike ad-hoc queries, reports are designed to be run regularly and shared with management. Two reports were generated for this system.

Q3-a) Borrowed Books Report — with Student Names

This report provides a complete picture of all borrowing activity. It joins the three core tables and computes a loan Status using a CASE expression — labelling records as Active, Returned, or Overdue based on the current date. This makes it immediately actionable for librarians.

```
SELECT br.BorrowID,
       s.StudentName,
       s.Program,
       b.Title AS BookTitle,
       b.Author,
       b.Category,
       br.BorrowDate,
       br.ReturnDate,
       CASE
         WHEN br.ReturnDate IS NULL THEN 'Active'
         WHEN br.ReturnDate >= CURDATE() THEN 'Returned'
         ELSE 'Overdue'
       END AS Status
FROM Borrow br
JOIN Students s ON br.StudentID = s.StudentID
JOIN Books b ON br.BookID = b.BookID
ORDER BY br.BorrowDate;
```

Report Output

BorrowID	StudentName	Program	BookTitle	BorrowDate	ReturnDate	Status
BR01	Ali Khan	BSCS	Database Fundamentals	01/03/26	10/03/26	Returned
BR02	John Lee	BSIT	Web Development	02/03/26	12/03/26	Returned

This report is accessible at GET /borrow and rendered in the React frontend's Borrow Report page with full pagination and a search bar. Each row is colour-coded by status for instant visual identification of overdue books.

Q3-b) Student Borrowing Summary Report

The Summary Report aggregates borrow data at the student level, counting how many books each student has borrowed. It uses a LEFT JOIN to ensure that students who have never borrowed a book still appear in the report with a count of zero — giving management a complete picture of engagement across the entire student body.

```
SELECT Students.StudentID,
Students.StudentName,
Students.Program,
Students.Year,
COUNT(Borrow.BorrowID) AS TotalBorrowed,
MAX(Borrow.BorrowDate) AS LastBorrowDate
FROM Students
LEFT JOIN Borrow ON Borrow.StudentID = Students.StudentID
GROUP BY Students.StudentID, Students.StudentName,
Students.Program, Students.Year
ORDER BY TotalBorrowed DESC;
```

Report Output

StudentID	StudentName	Program	Year	TotalBorrowed	LastBorrowDate
S101	Ali Khan	BSCS	2	1	2026-03-01
S103	John Lee	BSIT	3	1	2026-03-02
S102	Sara Ahmed	BBA	1	0	—

Notable insight: Sara Ahmed (BBA, Year 1) has zero borrow records. This is surfaced by the LEFT JOIN — without it, Sara would be completely absent from the report. The zero count signals either a catalogue gap for BBA students or a potential engagement opportunity for the library team.

Q3-c) How Reports Assist Management Decision-Making

Well-designed reports elevate a database from a passive data store into an active decision-support tool. The two reports generated for this system demonstrate how structured summaries directly enable smarter library management.

Report	Management Decision Enabled	Operational Benefit
Borrowed Books Report	Identify overdue loans requiring follow-up	Reduces book loss; improves return rates
Borrowed Books Report	Track which titles are in highest demand	Guides book acquisition and stock decisions
Student Summary Report	Spot students who never use the library	Enables targeted engagement programmes
Student Summary Report	Evaluate loan policy effectiveness	Supports policy reviews and limit adjustments
Both Reports	Demonstrate library ROI to university stakeholders	Justifies budget allocation for new books

Without automated reports, library administrators would need to manually count rows, calculate totals with a calculator, and compile summaries in spreadsheets — a process that is both time-consuming and prone to error. Automated database reports eliminate this entirely: a single click generates a fully up-to-date, accurate summary of the library's current state. Furthermore, reports create accountability — when management can see that Sara Ahmed has borrowed zero books, they can investigate whether the BBA curriculum recommends library resources, or whether the Business section needs expanding. This kind of insight is simply not achievable without structured reporting, making reports an indispensable management tool for any data-driven organisation.

Conclusion

This assignment demonstrated the end-to-end design and implementation of a University Library Management System using relational database principles. Three SQL queries were developed to retrieve meaningful data from multiple tables efficiently. Two data-entry forms were designed with validation to ensure accurate and user-friendly data input. Two management reports were generated to transform raw records into actionable insights.

Together, these components illustrate the three pillars of a practical database system: efficient data retrieval through queries, controlled data entry through forms, and strategic insight through reports. The system was implemented as a full-stack web application using MySQL for the database layer, Node.js and Express for the REST API backend, and React for the frontend interface — demonstrating that modern database systems are integrated components of live, user-facing applications rather than standalone tools.

SUMMARY OF DELIVERABLES:

Question 1 (8 Marks) — Three SQL queries designed and executed with results: Q1-a join query, Q1-b category filter, Q1-c date filter. Written analysis of query efficiency provided.

Question 2 (6 Marks) — Student entry form and book borrowing form designed with full validation rules documented. Explanation of how forms improve accuracy and usability.

Question 3 (6 Marks) — Borrowed books report and student summary report generated with complete SQL. Evaluation of reports as management decision-making tools provided.

University Library Management System — Database Systems Assignment — Week 11

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